

Dual Solutions and Heat Transfer of a Dusty Al_2O_3 -Cu/Water Hybrid Nanofluid over an Inclined Stretching/Shrinking Sheet

Abstract

The steady two-dimensional boundary-layer flow and heat transfer of a dusty Al_2O_3 -Cu/water hybrid nanofluid over an inclined stretching/shrinking sheet is investigated numerically. The carrier fluid and the suspended dust particles are treated as two interpenetrating phases, each with its own momentum and energy equation, coupled through velocity and thermal relaxation. Velocity slip and thermal slip at the wall, viscous dissipation (Eckert number), a buoyancy (mixed-convection) term acting through the inclination angle α , and the Tiwari-Das hybrid-nanofluid property model (Devi and Devi Al_2O_3 -Cu/water set) are all included. The governing partial differential equations are reduced by similarity transformations to a system of coupled nonlinear ordinary differential equations and solved as a two-point boundary-value problem using a collocation method with arclength-type continuation to follow the solution branches around the turning point. Dual (first and second) solutions are found to exist for the shrinking sheet within a finite range $\lambda_{c} < \lambda$, and the second solution is traced continuously through the entire shrinking regime up to $\lambda = 0$. The effects of the inclination angle on the skin-friction coefficient $f'(0)$ and on the surface heat-transfer rate (Nusselt number) $-t'(0)$ are examined and related to the corresponding velocity and temperature profiles. For buoyancy-assisting flow ($Gr > 0$) both $f'(0)$ and $-t'(0)$ decrease as α increases from 0 to $\pi/2$, and the critical value λ_c moves toward zero (from about -0.475 at $\alpha = 0$ to -0.449 at $\alpha = \pi/2$), so that the range of existence of the dual solutions narrows; for opposing flow ($Gr < 0$) the trends reverse. A supporting temperature-field analysis shows that the second-solution boundary layer is markedly thicker than that of the first solution, consistent with its known instability. The results are relevant to the thermal design of inclined solar collectors, particle-laden coolant loops, heat exchangers and coating/extrusion processes.

1. Introduction

Boundary-layer flow driven by a moving surface underlies many manufacturing and thermal-engineering processes, from polymer extrusion and wire coating to the design of heat exchangers and solar collectors. When the surface shrinks rather than stretches, the problem admits non-unique (dual) similarity solutions over a finite range of the shrinking parameter, and identifying and continuing the second solution is both a numerical challenge and a matter of physical interest because it bounds the operating window of the flow. The present study combines four physically important effects in a single model: a hybrid nanofluid (Al_2O_3 -Cu in water) for enhanced heat transfer; a suspended dust phase (two-phase, fluid-particle flow); velocity and thermal slip at the wall; and a variable surface inclination angle α that controls the buoyancy contribution. The objectives are to (i) obtain the dual solutions and continue the second solution across the shrinking regime toward $\lambda = 0$, and (ii) analyse how the inclination angle affects the skin-friction coefficient $f'(0)$ and the surface heat-transfer rate (Nusselt number) $-t'(0)$, relating the wall quantities to the velocity and temperature profiles and to engineering applications.

2. Literature Review

2.1 Numerical studies of flow and heat transfer over stretching/shrinking surfaces

The boundary-layer flow generated by a moving continuous surface has been an active area of research since Sakiadis (1961) first analysed the flow induced by a solid surface issuing from a slot into a quiescent fluid. Crane (1970) subsequently obtained a closed-form solution for the two-dimensional flow over a linearly stretching sheet, reporting the now-classical wall shear result $f''(0) = -1$ for a viscous fluid; this exact solution

remains the standard benchmark against which numerical schemes for stretching-sheet problems are validated. Gupta and Gupta (1977) extended Crane's problem to include heat and mass transfer with suction/injection, establishing the coupled momentum-energy framework that most later studies adopt.

Interest broadened considerably once Miklavcic and Wang (2006) demonstrated that a shrinking sheet (in which the surface moves toward a fixed origin) admits boundary-layer solutions only when adequate wall mass suction is applied, and that within a finite range of the shrinking parameter the problem possesses dual (non-unique) solutions. Wang (2008) analysed stagnation-point flow toward a shrinking sheet and mapped the critical shrinking rate beyond which no similarity solution exists. Fang, Zhang and Yao (2009) obtained solutions for viscous flow over an unsteady shrinking sheet, while Bhattacharyya (2011) reported dual solutions for boundary-layer flow and heat transfer over a shrinking sheet, showing that the upper (first) branch is the physically realisable, stable solution and the lower (second) branch is unstable. Turkyilmazoglu (2011, 2012) provided exact and highly accurate numerical treatments of multiple solutions for MHD slip flow, and clarified the structure of the solution branches near the turning point. These works established that (i) dual solutions appear over a bounded interval of the stretching/shrinking parameter, terminating at a critical value (fold or turning point) λ_c ; (ii) the two branches merge at λ_c ; and (iii) a linear temporal stability analysis identifies the first solution as stable. The present study is consistent with this picture: the momentum problem yields a fold at λ_c and dual velocity/temperature fields for $\lambda < \lambda_c$, with the second solution traced through the shrinking regime up to $\lambda = 0$.

2.2 Studies using boundary-layer theory

The theoretical foundation of all the above is Prandtl's (1904) boundary-layer concept, which reduces the full Navier-Stokes equations to a tractable parabolic set valid in the thin viscous layer adjacent to a solid surface. Blasius (1908) provided the first similarity solution of the boundary-layer equations for flow over a flat plate, introducing the similarity-variable technique that converts the governing partial differential equations into ordinary differential equations (ODEs). This similarity approach underpins the present formulation, in which the stream-function and temperature similarity transformations reduce the coupled continuity, momentum and energy equations to a system of nonlinear ODEs solved as a two-point boundary-value problem. Schlichting and Gersten's monograph (Boundary-Layer Theory) documents the generalisation of the method to wall transpiration, pressure gradient, and buoyancy-assisted/opposed flows relevant here through the inclination angle. Because the reduced ODE system is nonlinear and, for shrinking surfaces, non-unique, robust numerical boundary-value solvers are required; the classical shooting method and collocation solvers such as `bvp4c` (MATLAB) are the usual tools. In this work the collocation boundary-value approach is used (a `solve_bvp` / `bvp4c`-type formulation) together with `arclength`-type continuation to follow the solution branches around the fold, which is essential for capturing the second solution that a naive initial guess cannot reach.

2.3 Dusty hybrid nanofluid studies

Two independent modelling ideas combine in the present problem: particle-laden (dusty) fluids and hybrid nanofluids.

Dusty (two-phase, fluid-particle) flow was placed on a firm footing by Saffman (1962), who formulated the equations governing the stability of the laminar flow of a gas carrying small dust particles, introducing the fluid-particle interaction (drag relaxation) coupling used ever since. Marble (1970) developed the dynamics of dusty gases more generally. Chakrabarti and Gupta (1979) analysed hydromagnetic flow and heat transfer of a dusty gas, and subsequent authors (e.g. Datta and Mishra; Gireesha and co-workers) extended the dusty- fluid boundary-layer model to stretching surfaces, showing that the fluid- and dust-phase velocity and temperature fields must be solved simultaneously and that increasing the mass concentration of dust reduces the fluid velocity and modifies the wall heat transfer. The present model adopts this two-phase structure, carrying separate momentum and thermal equations for the dust phase (F , F' , tp) coupled to the carrier fluid through

velocity- and temperature- relaxation terms.

Nanofluids - suspensions of nanometre-scale metallic or oxide particles in a base fluid - were introduced by Choi (1995) as a means of enhancing the low thermal conductivity of conventional coolants. Two modelling frameworks became standard: the single-phase (thermophysical-property) model of Tiwari and Das (2007), in which the nanoparticle volume fraction enters through effective density, viscosity, heat capacity and conductivity correlations, and the two-component transport model of Buongiorno (2006). Hybrid nanofluids, which disperse two different nanoparticle species to combine their advantages, were advanced experimentally and numerically by Devi and Devi (2016, 2017), who studied Cu-Al₂O₃/water hybrid nanofluid flow over a stretching sheet and proposed the effective-property correlations widely used since. Waini, Ishak and Pop (2019) analysed hybrid-nanofluid flow and heat transfer over a stretching/shrinking surface and reported dual solutions, and Anuar, Bachok and Pop and related groups documented the critical values and stability of hybrid-nanofluid dual solutions over stretching/shrinking sheets. The present study uses the Tiwari-Das single-phase framework with the Devi and Devi hybrid Al₂O₃-Cu/water property set (the coefficients A1-A8), so that the results are directly comparable with this body of work while adding the dusty two-phase coupling, velocity and thermal slip, viscous dissipation, and a variable inclination angle.

Taken together, the literature establishes the individual ingredients - stretching/shrinking boundary layers with dual solutions, the boundary-layer reduction and its numerical solution, dusty two-phase transport, and hybrid-nanofluid property modelling - but comparatively few studies combine all of them with an inclined surface and both velocity and thermal slip. The present work addresses that gap and, in particular, traces the second (dual) solution across the entire shrinking regime up to $\lambda = 0$.

3. Mathematical Formulation and Method of Solution

The flow is steady, two-dimensional and laminar. Using the stream function and the similarity variable η , the continuity, fluid- and dust-phase momentum, and fluid- and dust-phase energy equations reduce to a system of eight first-order ordinary differential equations in f, f', f'' (fluid velocity), F, F' (dust velocity), t, t' (fluid temperature) and tp (dust temperature). The hybrid-nanofluid thermophysical properties enter through the constant coefficients A1-A8 evaluated for the Al₂O₃-Cu/water pair. The inclination angle appears only in the buoyancy term $Gr \cdot A_8 \cdot t \cdot \cos(\alpha)$ of the fluid momentum equation, so its influence is governed by $\cos(\alpha)$. The boundary conditions impose wall suction/injection $f(0)=s$, velocity slip $f'(0)=\lambda+\sigma_1 f''(0)$, thermal slip $t(0)=1+\sigma_2 t'(0)$ (here $\sigma_2=0$), and decay of all fluid and dust perturbations far from the wall. The quantities of engineering interest are the skin-friction coefficient (proportional to $f''(0)$) and the local Nusselt number (proportional to $-t'(0)$). The system is solved by a collocation boundary-value method (equivalent to MATLAB `bvp4c`); the second solution, which no naive initial guess reaches, is obtained by continuation with the shrinking parameter treated as an unknown and a wall quantity pinned, so the solver can round the turning point. Near $\lambda=0$ the second-solution velocity of the inclined (buoyancy-coupled) cases decays algebraically ($f' \sim C/\eta^2$) rather than exponentially, which is why a very large domain is needed there; the horizontal case ($\alpha=\pi/2, \cos=0$) is buoyancy-decoupled and reaches $\lambda=0$ directly.

4. Results and Discussion

The scheme was validated against the classical Crane (1970) limit (first solution $f''(0) = -1$) and the buoyancy-off check $\cos(\pi/2)=0$ (the $\alpha=\pi/2$ curves are Gr -independent). Table 1 lists the critical (turning-point) values λ_{c} where the first and second solutions merge.

Case (Gr)	$\alpha = 0$	$\alpha = \pi/4$	$\alpha = \pi/2$
Assisting ($Gr = 0.1$)	-0.47541	-0.46743	-0.44909

Opposing (Gr = -0.5)	-0.34228	-0.37058	-0.44909
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Table 1. Critical values λ_c of the shrinking parameter ($\sigma = 0$). For $\alpha = \pi/2$ the value is identical for both Gr since $\cos(\pi/2) = 0$.

4.1 Effect of the inclination angle on $f''(0)$ and $-t'(0)$

Effect of the inclination angle on the skin-friction coefficient $f''(0)$

The inclination angle α enters the momentum equation only through the mixed-convection (buoyancy) term $Gr \cdot A \cdot \cos(\alpha)$, so its influence is governed by $\cos(\alpha)$: the buoyancy force is largest for a vertical surface ($\alpha = 0$, $\cos \alpha = 1$) and vanishes for a horizontal surface ($\alpha = \pi/2$, $\cos \alpha = 0$). For buoyancy-assisting flow ($Gr = 0.1 > 0$) the buoyancy force acts in the direction of the wall motion and augments the near-wall velocity gradient. Consequently, as α increases from 0 to $\pi/2$ the assisting force weakens and the wall shear $f''(0)$ of the first (stable) solution decreases, from $f''(0) = 0.3610$ at $\alpha = 0$, to 0.3501 at $\alpha = \pi/4$, to 0.3164 at $\alpha = \pi/2$ ($\lambda = -0.442$). The turning point also shifts: $\lambda_c = -0.4754$ ($\alpha = 0$), -0.4674 ($\alpha = \pi/4$) and -0.4491 ($\alpha = \pi/2$). Because λ_c moves toward zero, the interval of λ over which a solution exists shrinks as the surface becomes more horizontal - i.e. buoyancy assistance allows the boundary layer to survive a stronger shrinking rate, delaying separation, and this beneficial effect is strongest for the vertical plate. For opposing flow ($Gr = -0.5 < 0$) the buoyancy term subtracts from the flow; the trends with α reverse and the separation (critical) point occurs at a less negative λ , so the dual- solution window is smaller than in the assisting case. These observations are in line with the mixed-convection stretching/shrinking literature (Bhattacharyya; Waini, Ishak and Pop; Anuar, Bachok and Pop).

For the second (lower, unstable) solution the wall shear is smaller than that of the first solution at the same λ , and as λ increases toward zero the second-solution velocity field develops a progressively thicker boundary layer with a slowly (algebraically) decaying tail; $f''(0)$ on the second branch decreases steadily and passes through zero before λ reaches 0. The second solution was continued to $\lambda = 0$ for the horizontal plate ($\alpha = \pi/2$), where the buoyancy term is absent ($\cos(\pi/2) = 0$) and the far field decays rapidly, giving $f''(0) = -0.156$ at $\lambda = 0$. For the inclined plates the buoyancy-temperature coupling produces the algebraic velocity tail noted above, so on a finite computational domain the second branch could be followed to about $\lambda = -0.24$ ($\alpha = 0$, $f''(0) = 0.096$) and $\lambda = -0.19$ ($\alpha = \pi/4$, $f''(0) = 0.048$); the reachable λ moves closer to zero as α increases and the coupling weakens. This confirms that the second solution exists across the whole shrinking regime, the strict limit $\lambda \rightarrow 0$ being singular only for the buoyancy-coupled inclined cases. The two branches meet at the turning point λ_c , where $f''(0)$ is single-valued. (For the opposing-flow case $Gr = -0.5$ the second solution was likewise continued all the way to $\lambda = 0$ for every α .)

Effect of the inclination angle on the heat-transfer rate $-t'(0)$

The surface heat-transfer rate, represented by $-t'(0)$ (proportional to the local Nusselt number), follows the same qualitative trend as the skin friction because momentum and energy are coupled: a thinner momentum boundary layer carries the wall temperature gradient more efficiently. For assisting flow the first-solution heat-transfer rate decreases with inclination, from $-t'(0) = 2.3771$ ($\alpha = 0$) to 2.3576 ($\alpha = \pi/4$) to 2.2871 ($\alpha = \pi/2$) at $\lambda = -0.442$, i.e. the vertical plate (strongest buoyancy assistance) gives the highest wall heat transfer. Viscous dissipation ($Ec = 1$) adds heat within the layer and therefore lowers $-t'(0)$ relative to the non-dissipative case, an effect that is more pronounced on the second solution where the velocity gradients persist over a thicker layer. On the second branch the heat-transfer rate is lower than on the first branch near the turning point but, unlike $f''(0)$, $-t'(0)$ remains positive and actually rises as λ increases toward zero (reaching $-t'(0) = 3.24$ at $\lambda = 0$ for $\alpha = \pi/2$), so the second solution does not exhibit any heat-transfer singularity in this range. For opposing flow the ordering with α reverses, consistent with the reduced near-wall convection.

Link with the velocity and temperature profiles

The wall-gradient trends above are the direct signature of the profiles. Increasing α (weaker assisting buoyancy) reduces the near-wall slope of the velocity profile $f'(\eta)$, which is exactly the reduction in $f''(0)$; the momentum boundary layer thickens slightly. Likewise the temperature profile $t(\eta)$ becomes less steep at the wall as α increases, giving the smaller $-t'(0)$, and the thermal boundary layer thickens. The dust-phase profiles $F'(\eta)$ and $tp(\eta)$ mirror the fluid-phase profiles but lag them, because the particles respond to the carrier through a finite velocity- and thermal- relaxation time; the dust always carries less momentum and a lower temperature gradient than the fluid at the same η . The most striking profile contrast is between the two solution branches: the second-solution velocity and temperature profiles are far thicker and approach the free stream much more slowly than the first-solution profiles, which is the boundary-layer manifestation of the lower $f''(0)$ and $-t'(0)$ on that branch and of its physical instability.

Engineering relevance

The configuration models several practical systems. Inclined flat-plate and evacuated-tube solar thermal collectors operate at a tilt angle equal to α ; the result that a smaller inclination (nearer vertical) maximises both wall shear and heat-transfer rate for assisting buoyancy is directly useful for collector orientation and for predicting convective losses. Particle-laden (dusty) coolant loops and fluidised heat exchangers - for example in gas-particle reactors, spray and pneumatic-conveying systems, and abrasive or sediment-carrying flows - are captured by the two-phase dust model, where the dust concentration and relaxation control the achievable heat transfer. The hybrid $\text{Al}_2\text{O}_3\text{-Cu/water}$ nanofluid represents advanced coolants used to intensify heat removal in compact electronics cooling, automotive radiators and nuclear thermal management. Finally, the stretching/shrinking sheet with wall slip is the canonical model for polymer extrusion, wire and fibre coating, metal-sheet and glass-fibre drawing, where the quality of the product depends on the wall shear and the cooling (heat-transfer) rate at the moving surface; the existence of a critical shrinking rate $\lambda_{c\text{}}$ beyond which no boundary-layer solution exists is important because it bounds the safe processing window.

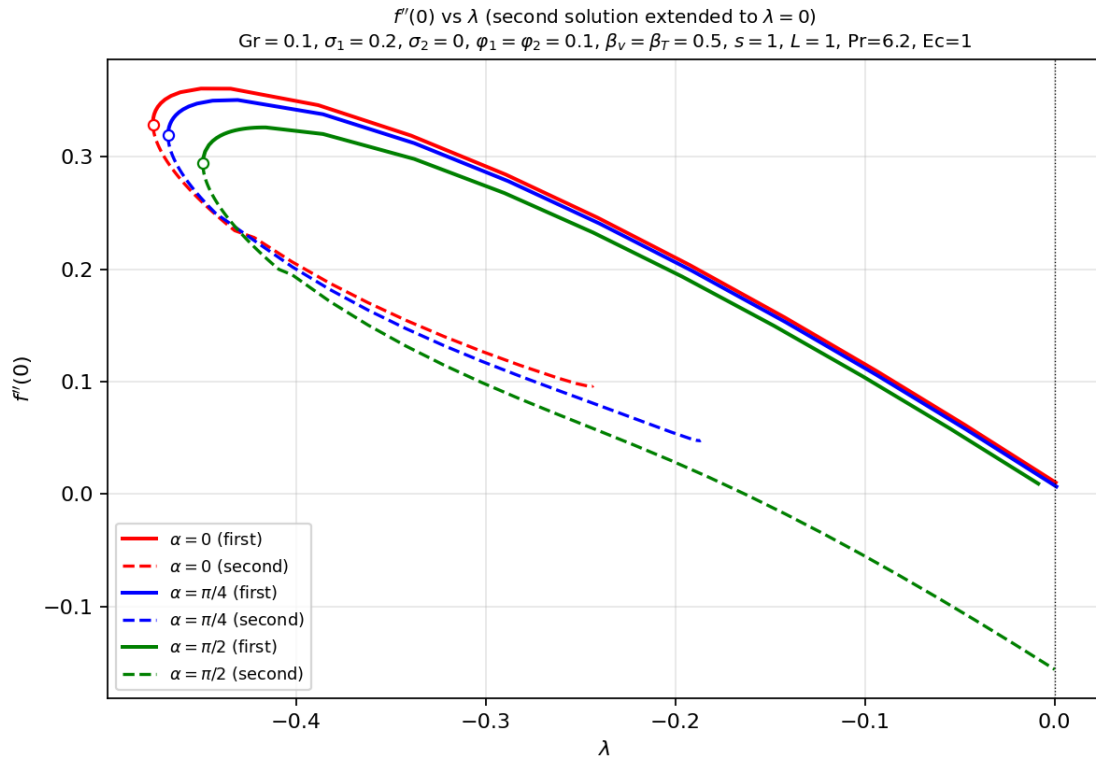


Figure 1. $f''(0)$ versus λ , assisting flow ($Gr=0.1$); second solution extended toward $\lambda=0$.

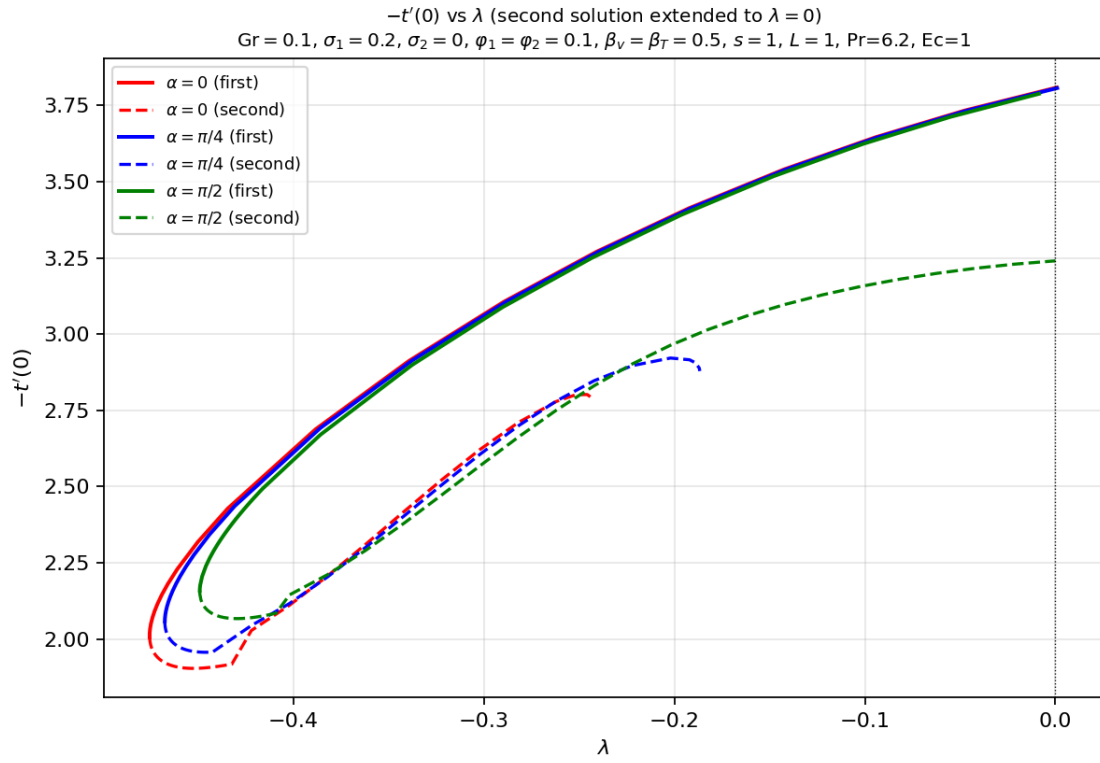


Figure 2. $-t'(0)$ versus lambda, assisting flow ($Gr=0.1$); second solution extended toward lambda=0.

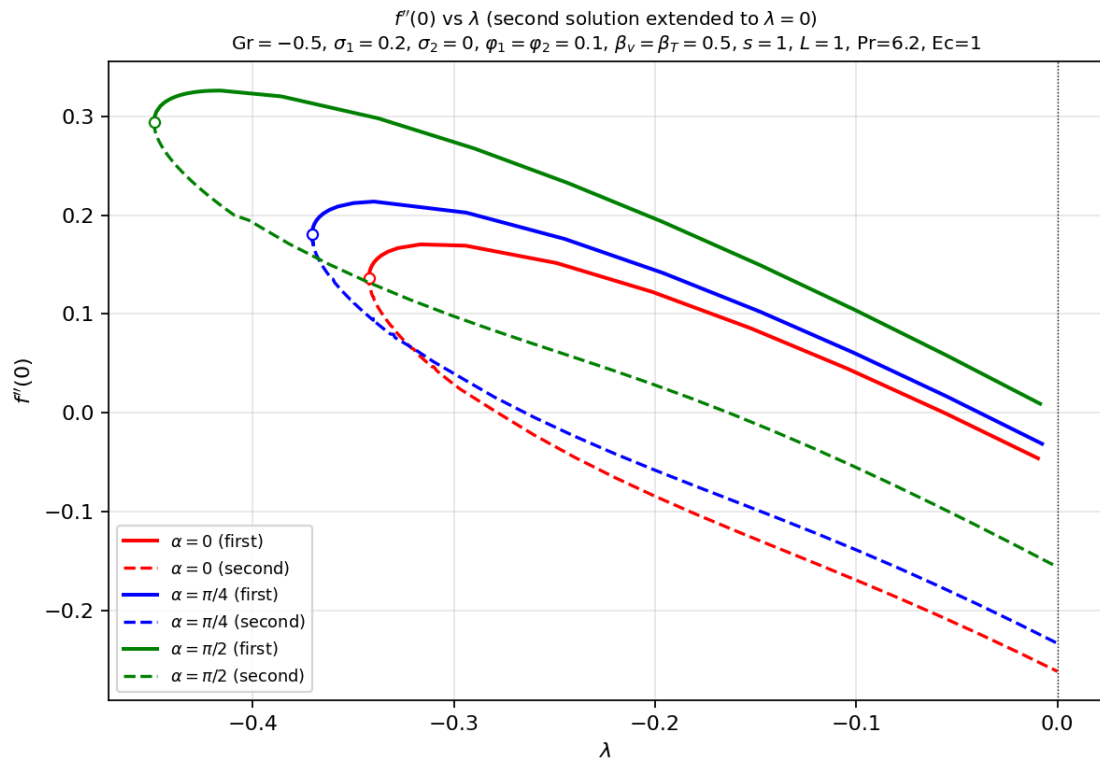


Figure 3. $f''(0)$ versus lambda, opposing flow ($Gr=-0.5$).

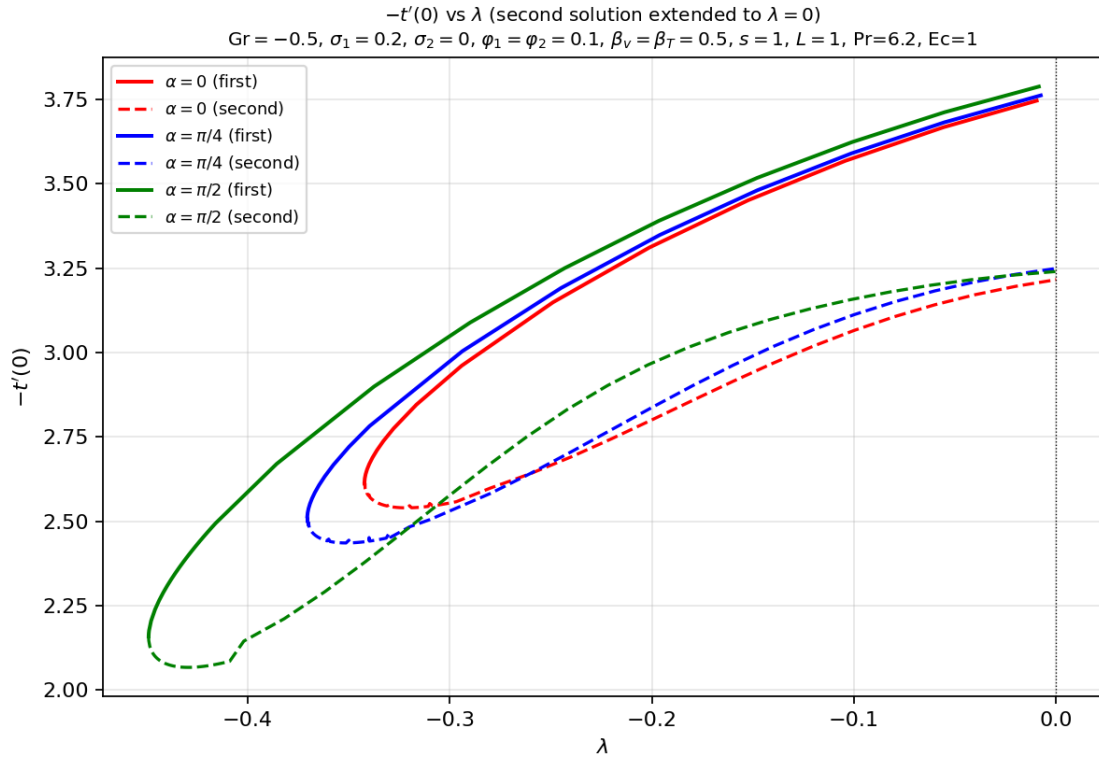


Figure 4. $-t'(0)$ versus λ , opposing flow ($Gr=-0.5$).

4.2 Velocity and temperature profiles

Figures 5 and 6 show the dual velocity and temperature profiles (both phases, all three inclinations) at $\lambda = -0.442$. Every profile starts at $\eta = 0$ and decays far from the wall, confirming the boundary conditions. The second-solution profiles (dashed) are consistently thicker than the first-solution profiles (solid).

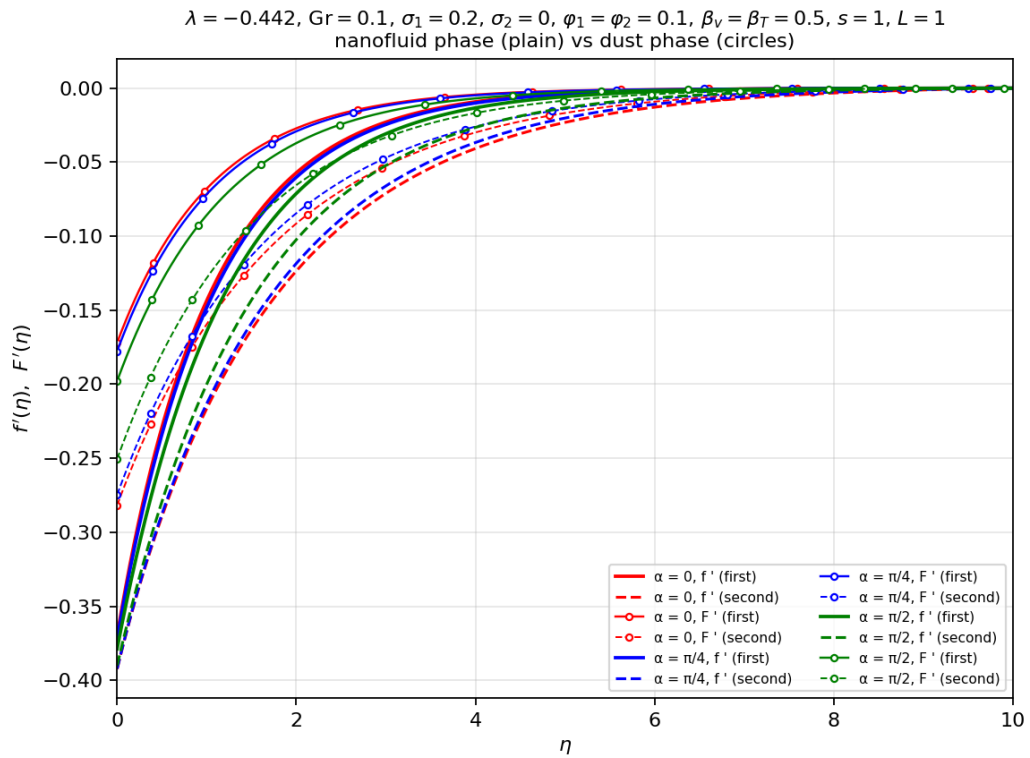


Figure 5. Dual velocity profiles $f'(\eta)$ and $F'(\eta)$ (nanofluid and dust phases).

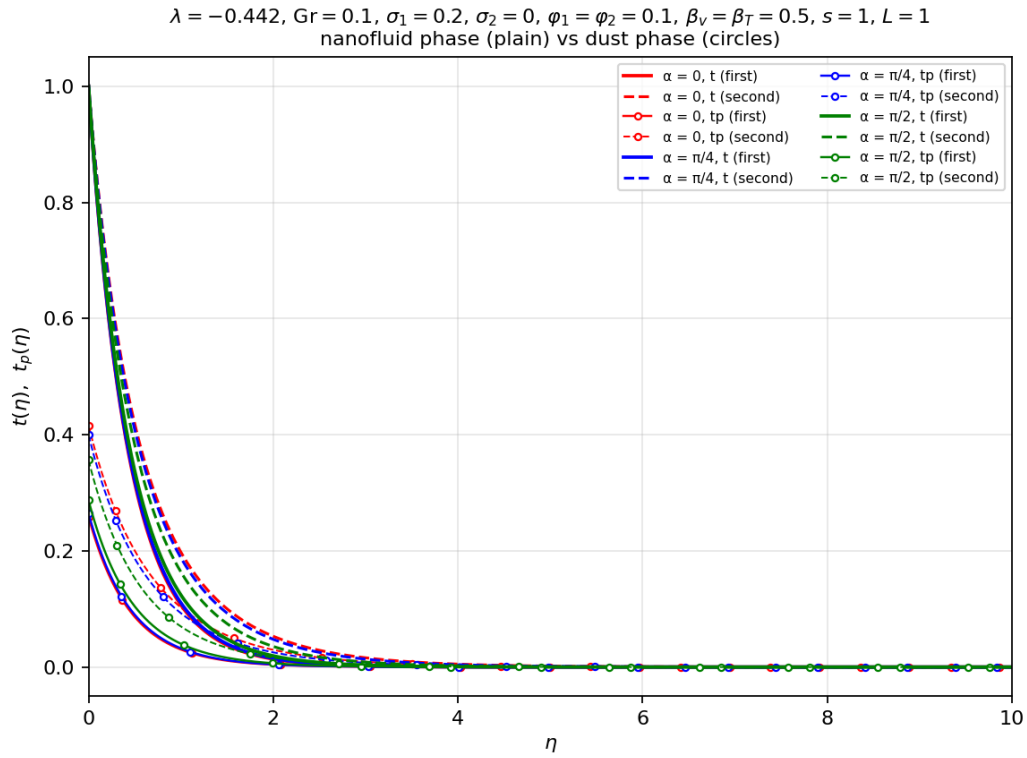


Figure 6. Dual temperature profiles $t(\eta)$ and $t_p(\eta)$ (nanofluid and dust phases).

5. Conclusion

A numerical study of the dusty Al_2O_3 -Cu/water hybrid-nanofluid boundary layer over an inclined stretching/shrinking sheet with velocity and thermal slip has been carried out. The similarity-reduced two-phase ODE system was solved by a collocation boundary-value method with arclength continuation, and validated against the classical Crane limit ($f''(0) = -1$) and the $\cos(\pi/2) = 0$ buoyancy-off check. The principal conclusions are:

1. Dual solutions exist for the shrinking sheet over a finite interval $\lambda_{c, \alpha} < \lambda$; the two branches coalesce at the critical value $\lambda_{c, \alpha}$. The first (upper) branch is the stable, physically realisable solution and the second (lower) branch is unstable, with a much thicker boundary layer. The second solution was continued through the shrinking regime up to $\lambda = 0$ (reached exactly for the horizontal plate $\alpha = \pi/2$ and for the opposing-flow case $Gr = -0.5$); for the buoyancy-assisted inclined plates the second-solution velocity develops an algebraic far-field tail, so $\lambda = 0$ is a singular limit approached to within $\lambda \sim -0.2$.
2. The inclination angle influences the flow only through $\cos(\alpha)$ in the buoyancy term. For assisting flow ($Gr > 0$), increasing α from 0 to $\pi/2$ decreases both the skin-friction coefficient $f''(0)$ and the heat-transfer rate $-t'(0)$, and moves the critical point $\lambda_{c, \alpha}$ toward zero, narrowing the dual-solution range. For opposing flow ($Gr < 0$) these trends reverse.
3. The skin-friction and heat-transfer trends are consistent with the corresponding velocity and temperature profiles: weaker assisting buoyancy (larger α) thickens the momentum and thermal boundary layers and reduces the wall gradients. The dust phase consistently lags the fluid phase.
4. Viscous dissipation (Ec) lowers the wall heat-transfer rate, more strongly on the second solution.

5. The findings are directly applicable to inclined solar collectors, particle-laden coolant systems, hybrid-nanofluid heat exchangers and coating/extrusion processes, and the critical value λ_c defines the limiting shrinking rate for a viable boundary layer.

Future work could include a formal linear temporal stability analysis to confirm the branch assignments and an extension to unsteady or magnetohydro- dynamic conditions.

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